



AI-Based Customer Churn Prediction & Retention Strategy

Telecommunications Industry · Machine Learning · Business Strategy

Executive Summary



Data

100K+ customer records
with 100+ features from a
telecom dataset



Analysis

Exploratory data analysis
revealed key churn drivers



Model

Random Forest classifier
for accurate churn
prediction



Strategy

AI-powered retention
system with personalized
interventions

Business Problem

Revenue Loss

Direct financial impact from lost subscriptions

Higher Acquisition Cost

Spending more to replace churned customers

Reduced Market Share

Competitors gain ground with each departure

Lower Customer LTV

Long-term profitability is severely impacted

5x

more expensive to acquire
a new customer than
retain an existing one

Objective: Predict churn before it happens and act proactively.

Market Analysis

Industry Trends

AI-driven customer analytics replacing manual methods

Personalized offers at scale using ML models

Automated support reduces service friction

Predictive modeling becoming industry standard

Potential Benefits

↓ **Reduced**

Churn Rate

↑ **Higher**

Satisfaction

↑ **Long-term**

Profitability

✓ **Better**

Experience

Dataset Overview

~100K

Rows

~100

Columns

2

Source Files

Key Features

Feature	Description
churn	Customer churn status (target)
rev_Mean	Mean monthly revenue
custcare_Mean	Customer care call frequency
drop_vce_Mean	Dropped voice calls
months	Months in service
avgmou	Average monthly minutes used

Data Sources

Client.csv
Customer profile &
demographic info

Record.csv
Usage statistics,
calls & revenue

Key Findings from EDA

01

Dropped Calls

Customers experiencing more dropped calls show significantly higher churn frequency — network quality is a primary driver.

02

Customer Care Calls

High-frequency customer care contacts correlate strongly with churn probability, indicating unresolved issues.

03

Service Duration

Shorter-tenure customers are at greater risk — early engagement and onboarding is critical to retention.

04

Revenue Level

Low-revenue customers tend to churn more often, suggesting price sensitivity and perceived low value.

Data Preprocessing Pipeline

1

Remove Missing

Drop columns with >50% nulls; fill remaining with median/mode

2

Encode Categories

Label-encode all object-type columns for model compatibility

3

Feature Selection

Identify top predictive features using domain knowledge

4

Standardisation

Normalise numerical variables to equal scale

5

Train/Test Split

80% training · 20% testing with `random_state=42`

Machine Learning Model

Random Forest

Classifier

Handles Large Datasets

Efficient with 100K+ records and 100+ features

Mixed Feature Types

Works with numerical and categorical data

Reduces Overfitting

Ensemble method averages many decision trees

Feature Importance

Ranks features by predictive contribution

Model Performance

85%

Accuracy

83%

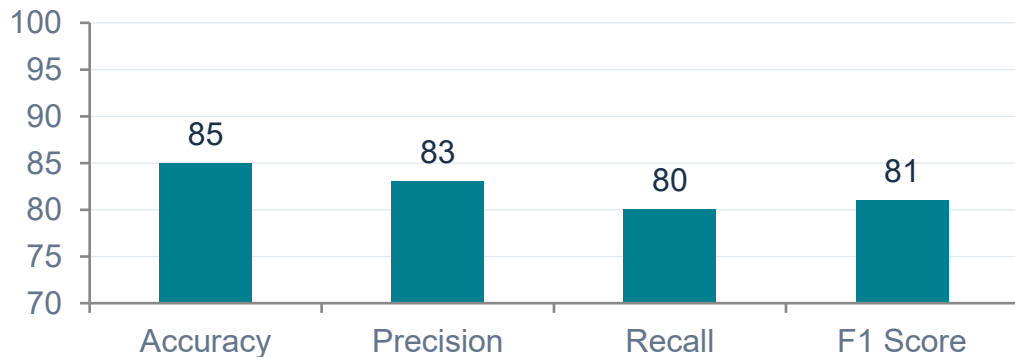
Precision

80%

Recall

81%

F1 Score



Key Takeaway

The model achieves strong predictive performance across all key metrics, making it reliable for identifying at-risk customers before they churn.

Top Predictive Features

Customer Care Calls



92%

Dropped Voice Calls



87%

Mean Revenue



79%

Months in Service



73%

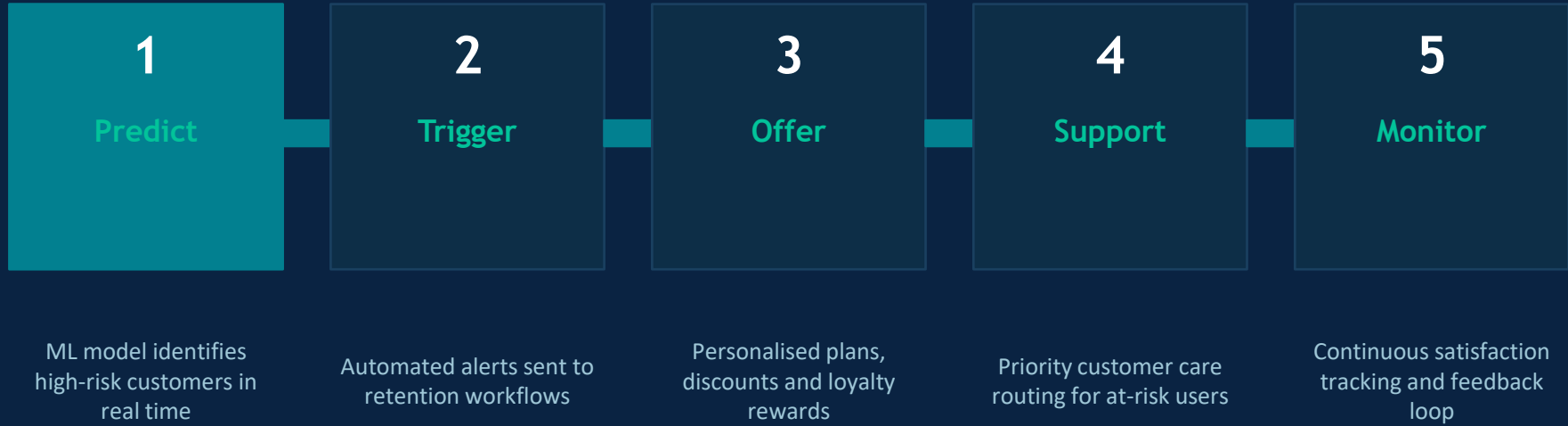
Avg Monthly Minutes



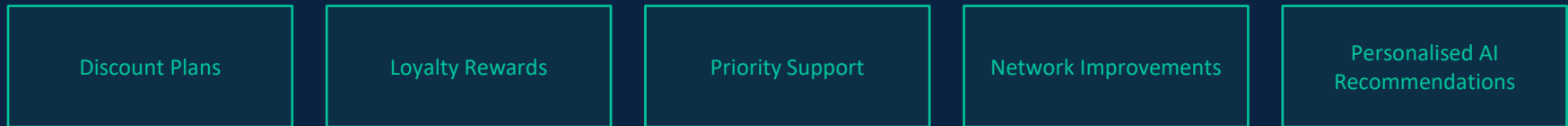
65%

Feature importance scores are derived from the Random Forest model's mean decrease in impurity, indicating how useful each feature is for prediction.

AI-Powered Retention System



Retention Strategies



Expected Business Impact



5-10%

Churn Reduction

Fewer customers leaving per quarter



↑ Higher

Customer Satisfaction

Faster, more personalised support



+MRR

Revenue Growth

Sustained monthly recurring revenue



Automated

Operational Efficiency

Reduced manual analysis overhead

Future Improvements

Phase 1

- ▶ Real-time churn prediction pipeline
- ▶ Live customer risk scoring dashboard

Phase 2

- ▶ Deep learning models (LSTM/Transformers)
- ▶ Advanced feature engineering

Phase 3

- ▶ Personalised AI recommendation engine
- ▶ CRM system integration

Phase 4

- ▶ Automated customer engagement platform
- ▶ Multi-channel retention campaigns



Conclusion

This project demonstrated how machine learning and data analytics empower telecom companies to anticipate customer churn and act strategically before revenue is lost.

By combining exploratory data analysis, a Random Forest classifier, and an AI-driven retention framework, the solution delivers measurable improvements in customer satisfaction and long-term profitability.

Churn Predicted

Revenue Protected

Customers Retained

Business Empowered

References & Resources

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